

# POLYNOMIAL REGRESSION

Machine Learning Assignment 1 | Roll number: BT2024191

## 1. Problem and dataset preparation

This project uses polynomial regression for two separate prediction problems. Q1 predicts the Net Power Score of a steam turbine from six operating parameters. Q2 predicts the Thermal Anomaly Score of a geothermal reservoir from three spatial coordinates. A separate model is fitted for each problem using the same Python program.

| Dataset   | Inputs | Labelled rows | Test rows | Degree range |
|-----------|--------|---------------|-----------|--------------|
| Q1 / var1 | 6      | 1,000         | 1,000     | 1-10         |
| Q2 / var2 | 3      | 1,000         | 1,000     | 1-20         |

The supplied training and test files are kept separate. The target column  $y$  is used only from the labelled training files. Model selection uses the training data, and predictions on the test rows retain their original order. Q2 contains 996 distinct input locations across 1,000 labelled rows; repeated locations are kept together during validation.

## 2. Polynomial features and standardisation

Each input column is standardised using its fitting-fold mean and standard deviation. All polynomial terms up to the chosen total degree, including interaction terms, are then generated and standardised again. For example,  $x_1$  squared times  $x_2$  is a degree-3 term. A separate intercept is fitted, so the expansion does not include a constant column.

Preprocessing is fitted again within every training fold. Validation rows are transformed with those fitted scalers, without contributing to their statistics. This is especially useful for high-degree terms, whose numerical scales can differ considerably. The final test inputs use the preprocessing fitted on all labelled rows.

## 3. Regularisation and search space

Ridge uses an L2 penalty to shrink coefficients, while Lasso uses an L1 penalty and can set coefficients to zero. Elastic Net combines both penalties. Alpha controls the penalty strength, and the Elastic Net L1 ratio controls the balance between L1 and L2. These settings are selected using validation MSE.

| Model       | Alpha values                               | L1 ratio      |
|-------------|--------------------------------------------|---------------|
| Ridge       | 0.0001, 0.001, 0.01, 0.1, 1, 10, 100, 1000 | 0             |
| Lasso       | 1, 0.1, 0.01, 0.001                        | 1             |
| Elastic Net | 1, 0.1, 0.01, 0.001                        | 0.2, 0.5, 0.8 |

Every permitted degree is screened with 24 configurations: 8 Ridge, 4 Lasso and 12 Elastic Net. A larger degree is not automatically better, and the error curve need not be convex. The chosen model is the best eligible configuration in this finite search, rather than a guaranteed global optimum.

## 4. Validation and model selection

Five-fold nested cross-validation separates hyperparameter selection from evaluation. For each outer fold, model selection uses only the remaining training rows. The selected model is then fitted on that outer training partition and evaluated on its held-out rows. Outer scores are not used to choose the final configuration.

| Stage                | Q1                                                                         | Q2                                                                                               |
|----------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Outer evaluation     | 800 fitting / 200 held out                                                 | Approximately 800 fitting / 200 held out                                                         |
| Inner screening      | 3 folds: approximately 533 fitting / 267 validation                        | Same 3-fold screening                                                                            |
| Additional selection | Lowest eligible screening MSE                                              | 6 shortlisted configurations; 5 folds repeated twice, approximately 640 fitting / 160 validation |
| Final model          | Selection repeated on all 1,000 labelled rows; winner refitted on all rows | Same approach, including repeated shortlist validation                                           |

There is no fixed 60/20/20 split. Each labelled observation is held out once in outer CV and is used for fitting in the other outer folds. The supplied test set remains separate throughout. Grouped splitting keeps identical input vectors on the same side of every split. The outer seed is 43 and the screening seed is 42.

### Q2: repeated inner validation

Within each training partition, the screen identifies the best two distinct degrees from each model family, retaining the best eligible penalty settings for each degree. These six configurations are then ranked using five grouped folds repeated twice, with seeds 142 and 143. The configuration with the smallest eligible mean MSE is selected. This checks the shortlist across more splits; its score is still a selection score because the shortlist was derived from the same training partition.

### Convergence and implementation checks

Sparse fits use a dual-gap check, with tolerance scaled to the centred target variance. A candidate is excluded if any relevant inner fit fails this check. Failed fits remain in the search records. Ridge uses a matrix-based solution, while sparse models use a warm-start regularisation path. Numerical checks compare solutions with independent reference solvers and check train-only scaling and duplicate isolation.

| Search work across 5 outer folds + final selection | Q1             | Q2       |
|----------------------------------------------------|----------------|----------|
| Screening fits                                     | 4,320          | 8,640    |
| Nonconvergent screening fits                       | 329            | 933      |
| Repeated-validation fits / nonconvergent fits      | Not applicable | 360 / 11 |

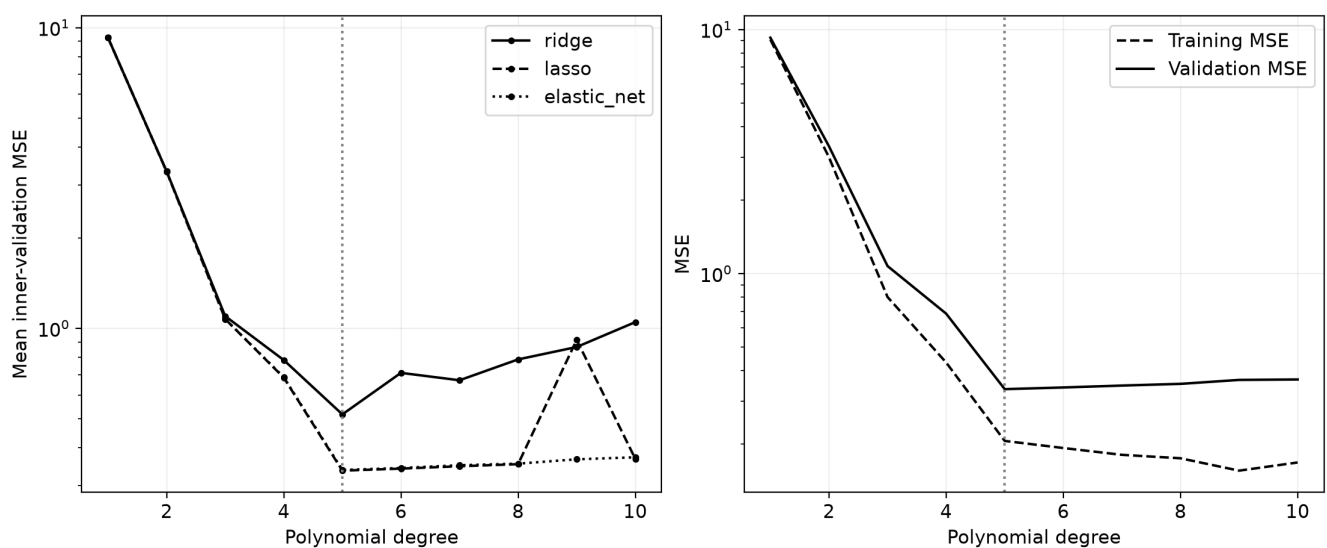
### Evaluation measures

MSE is the mean of (actual  $y$  - predicted  $y$ ) squared; lower values are better. R-squared is 1 minus the ratio of squared prediction error to total squared variation around the mean target. Pooled outer metrics are calculated from all 1,000 held-out predictions. Training MSE and full-data inner-selection MSE are reported separately, since neither is an independent test score.

## 5. Q1: Steam turbine prediction

| Result                                  | Value                        |
|-----------------------------------------|------------------------------|
| Selected model                          | Degree 5 Lasso, alpha = 0.01 |
| Polynomial terms / nonzero coefficients | 461 / 120                    |
| Pooled outer-validation MSE             | 0.315281                     |
| Pooled outer-validation R-squared       | 0.970134                     |
| Full-data inner-selection MSE           | 0.336027                     |
| Final-model training MSE                | 0.229104                     |

All five outer folds selected degree-5 Lasso with alpha 0.01, matching the final full-data selection. Of the 461 polynomial terms, 120 have nonzero coefficients. The full-data validation curve reaches its lowest eligible score at degree 5, while larger degrees do not improve it. This supports a regularised model with a moderate degree.



**Figure 1.** Full-data three-fold screening MSE by degree. Left: best eligible setting per family. Right: training and validation errors for the best eligible screening setting at each degree. MSE is on a logarithmic scale. The vertical line marks the final degree.

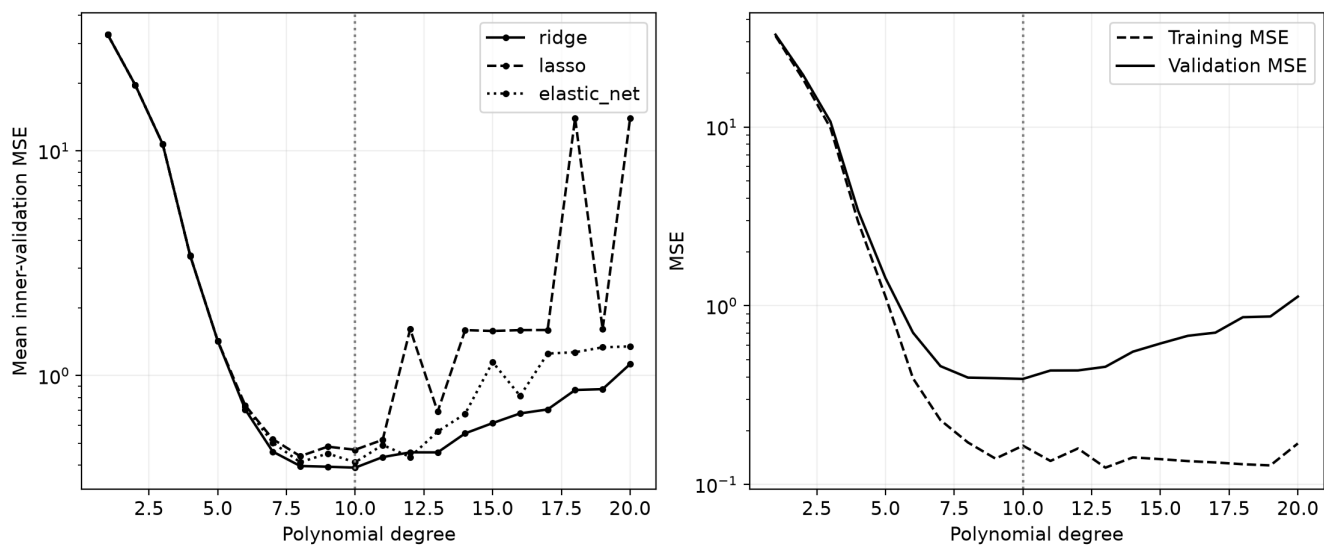
| Outer fold | Model | Degree | Alpha | Held-out MSE |
|------------|-------|--------|-------|--------------|
| 1          | lasso | 5      | 0.01  | 0.321445     |
| 2          | lasso | 5      | 0.01  | 0.266525     |
| 3          | lasso | 5      | 0.01  | 0.320801     |
| 4          | lasso | 5      | 0.01  | 0.395805     |
| 5          | lasso | 5      | 0.01  | 0.271830     |

The agreement across folds is encouraging, but it does not establish that overfitting is absent. Training error is lower than outer-validation error, so the held-out estimate is the more useful performance measure.

## 6. Q2: Thermal reservoir prediction

| Result                                  | Value                      |
|-----------------------------------------|----------------------------|
| Selected model                          | Degree 10 Ridge, alpha = 1 |
| Polynomial terms / nonzero coefficients | 285 / 285                  |
| Pooled outer-validation MSE             | 0.308099                   |
| Pooled outer-validation R-squared       | 0.993064                   |
| Full-data inner-selection MSE           | 0.283795                   |
| Final-model training MSE                | 0.180240                   |

Repeated validation on the full labelled dataset selected degree-10 Ridge with alpha 1. All 285 polynomial coefficients are retained. Outer-fold choices vary across model families and degrees, showing that selection is sensitive to the training split. The selected Ridge alpha lies within the searched range.

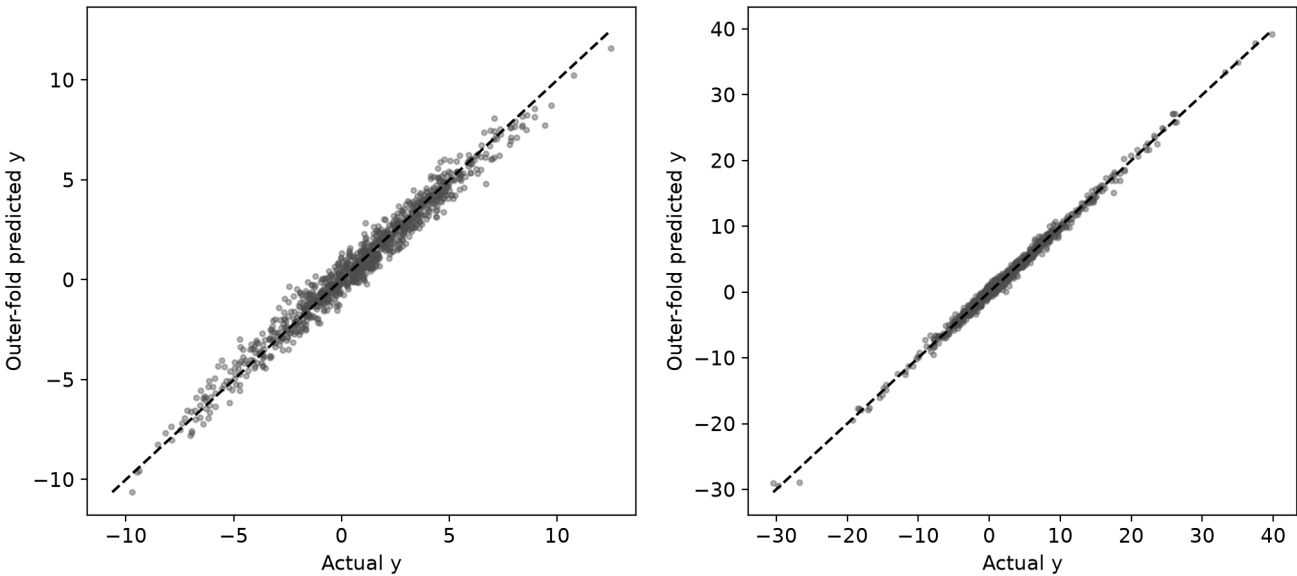


**Figure 2.** Full-data three-fold screening MSE by degree. Left: best eligible setting per family. Right: training and validation errors for the best eligible screening setting at each degree. MSE is on a logarithmic scale. The vertical line marks the final degree. For Q2, this degree is chosen after repeated validation; the plotted curves are screening scores.

| Outer fold | Model       | Degree | Alpha | Held-out MSE |
|------------|-------------|--------|-------|--------------|
| 1          | ridge       | 10     | 1     | 0.306633     |
| 2          | lasso       | 9      | 0.001 | 0.438271     |
| 3          | ridge       | 10     | 1     | 0.217187     |
| 4          | elastic net | 10     | 0.001 | 0.306490     |
| 5          | ridge       | 11     | 1     | 0.270628     |

Held-out fold sizes are 200, 201, 199, 201 and 199. Their unweighted mean MSE is 0.307842, while pooled MSE is 0.308099. The fold MSE standard deviation is 0.081575; this describes variation across folds and is not a confidence interval.

# 7. Final findings and predictions



**Figure 3.** Outer-fold predicted versus actual targets: Q1 on the left and Q2 on the right. Each point was predicted without using that observation or its duplicate input location for fitting. Points close to the diagonal indicate close agreement.

## Final model comparison

| Measure                              | Q1             | Q2              |
|--------------------------------------|----------------|-----------------|
| Final model                          | Degree-5 Lasso | Degree-10 Ridge |
| Outer-validation MSE                 | 0.315281       | 0.308099        |
| Outer-validation R-squared           | 0.970134       | 0.993064        |
| Labelled rows used for final fitting | 1,000          | 1,000           |
| Test predictions generated           | 1,000          | 1,000           |

## Interpretation and limitations

The chosen degrees balance fit and validation error within the tested search. Lasso gives a sparse Q1 model, while Ridge retains all terms for Q2. Regularisation, grouped splitting and fold-specific preprocessing reduce overfitting risk. Q1 has consistent outer-fold selections; Q2 has greater selection and error variation.

Convergence exclusions affect the available high-degree comparisons, and some sparse penalties lie at the lower search boundary.

## Final fitting and reproducibility

After full-data selection, each winning model is fitted on all 1,000 labelled rows. The saved models produce BT2024191\_pred\_var1.csv and BT2024191\_pred\_var2.csv, each with one y column and 1,000 finite predictions in the original test order. Reloaded-model predictions match the saved CSV values.

The shared main.py contains preprocessing, degree and penalty search, nested evaluation, fitting, inference and plotting. Candidate scores, convergence records, fold assignments, settings and package versions are retained alongside the saved models. Verification covers prediction sizes, model reloading, fold separation and recomputation of the pooled metrics. This keeps the final results traceable to the training procedure.